Rahul Pandey

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Engineering Manager & Architecture Leader | Python • System Design • Distributed Systems | 10+ Years

PROFESSIONAL SUMMARY

Engineering leader who architects and delivers enterprise-scale platforms across fintech, blockchain, and cloud-native systems. Built production infrastructure serving 100K+ users with 99.99% availability, including cross-border payment platforms and multi-tenant wallet ecosystems. Translates ambiguous business requirements into durable technical architectures, leads teams of 40+ engineers, and establishes engineering standards that reduce deployment time from 4 hours to 20 minutes. Strong depth in Python, React, AWS, and distributed system design.

KEY ACHIEVEMENTS

- Architected enterprise wallet infrastructure serving 50+ clients with 100,000+ active wallets, achieving 99.9% uptime and zero critical security incidents
- Led engineering organization growth from 5 to 40+ professionals while securing $5M+ in client contracts and reducing attrition to 8%
- Designed cross-chain bridge infrastructure securing 50,000+ transactions worth $25M+ with zero security incidents through comprehensive audits
- Authored 3 Hedera Improvement Proposals (HIPs) accepted into core protocol, receiving 'Best Web3 Innovation' award at Blockchain Summit 2023

SKILLS

Programming Languages: Python, JavaScript, TypeScript, Node.js, Rust, Solidity, Golang, Java

Frontend Frameworks: React, Next.js, Vue.js, Angular, Express.js, NestJS

Backend Frameworks: Django, Flask, Spring Boot

Cloud & Infrastructure: AWS (Solutions Architect Professional), Azure, Docker, Kubernetes, CI/CD, Terraform, Serverless, Lambda

Databases & Storage: PostgreSQL, MySQL, MongoDB, Redis, Cassandra, DynamoDB, IPFS, Arweave

System Design: Microservices, API Design, Distributed Systems, Technical Strategy, Architecture Patterns, NFR Definition

Security & Compliance: Smart Contract Auditing, SOC2, ISO 27001, KYC/AML, Privacy & Governance, Security Best Practices

Leadership: Team Building, Stakeholder Management, Agile/Scrum, Technical Discovery, Solution Design, Code Review

PROFESSIONAL EXPERIENCE

Senior Principal Engineer | Wissen Technology | December 2025 – Present

- Lead architecture and delivery of large-scale cross-border payment platforms used by international banks and payment processors

- Manage engineering team of 12 backend, frontend, DevOps, and QA engineers delivering mission-critical systems with 99.99% availability
- Define technical strategy and influence architecture across teams for fintech and payments practice
- Translate ambiguous business requirements into scalable technical architectures and design documents
- Review critical designs and codebases, establish engineering standards for security, scalability, and maintainability

Engineering Manager | Kalp Studio | March 2025 – October 2025

- Led cross-functional engineering teams of 15+ developers, architects, and QA professionals, increasing delivery velocity by 40% and reducing bug rates by 35%
- Architected enterprise-grade multi-tenant wallet infrastructure (Kalp WaaS) serving 50+ clients with 100,000+ active wallets, featuring HSM-backed key management and SOC2/ISO 27001 compliance
- Implemented scalable microservices architecture using Kubernetes, supporting 10,000+ concurrent users with sub-100ms transaction latency
- Directed end-to-end development of cross-platform mobile wallet (iOS/Android) with 25,000+ downloads and 4.8/5 star rating
- Spearheaded browser extension development for Chrome, Firefox, and Brave with 50,000+ active users, integrating with 500+ DeFi protocols
- Designed comprehensive DevOps pipeline with automated testing and blue-green deployment, reducing deployment time from 4 hours to 20 minutes
- Led migration from monolithic to microservices architecture, reducing cloud infrastructure costs by 30%
- Established engineering best practices for prompt design, model integration, evaluation, and observability

Chief Technology Officer | Web23 | March 2021 – March 2025

- Orchestrated company-wide technical vision and engineering standards, supporting 300% revenue growth over 3 years
- Built and mentored engineering organization of 40+ professionals across development, DevOps, QA, and security teams, reducing attrition to 8%
- Launched 20+ Web3 projects across Ethereum, Solana, Cardano, and Hedera, receiving 'Best Web3 Innovation' award at Blockchain Summit 2023
- Authored 3 Hedera Improvement Proposals (HIPs) accepted and integrated into core Hedera protocol
- Secured $500K organizational grant from Urbit Foundation as Identity and Journal Lead
- Architected NFT marketplace platform processing $10M+ in transaction volume with gasless transactions and multi-chain support
- Created blockchain-based real estate tokenization platform managing $50M+ in property value with SEC-compliant security token framework
- Developed Cardano Developer Toolkit with 10,000+ GitHub stars, including Plutus smart contract templates and wallet integration libraries
- Designed cross-chain bridge infrastructure enabling asset transfer between Ethereum, BSC, Polygon, and Avalanche, securing 50,000+ transactions worth $25M+ with zero security incidents
- Built DAO framework on Hedera managing $2M+ treasury with 5,000+ active participants
- Implemented blockchain donation tracking system for 15+ nonprofits, processing $5M+ with full audit trail
- Delivered Web3 domain resolution browser extension with 100,000+ installations
- Designed enterprise-grade AWS cloud infrastructure using Lambda, Cognito, CloudWatch, S3, and RDS
- Established CI/CD pipelines using GitHub Actions and Jenkins, achieving 50+ daily deployments
- Led US coupon industry platform (TheCouponBureau) handling 10M+ API calls daily using Node.js microservices and Redis caching

- Partnered with product, design, and business leaders to translate strategic objectives into scalable technical solutions
- Conducted technical discovery and solutioning workshops for high-impact initiatives

Blockchain Engineer | InfoKnob | November 2018 – March 2021

- Developed and deployed 15+ POCs and MVPs across Ethereum, Hyperledger Fabric, Solana, and Terra, converting 70% to production systems
- Built pharmaceutical supply chain tracking system ensuring drug authenticity across 500+ distribution points, reducing counterfeit incidents by 95%
- Designed DeFi lending protocol on Ethereum with $1M+ TVL, achieving 40% reduction in transaction costs through gas optimization
- Created NFT marketplace facilitating 10,000+ transactions with ERC-721/ERC-1155 standards and IPFS integration
- Developed Rust-based consensus algorithms optimizing transaction throughput to 5,000+ TPS on private networks
- Managed 100TB+ of immutable data using IPFS and Arweave distributed file systems
- Conducted POCs to validate design and technology choices before production implementation

Senior Software Engineer | Expedia Group | May 2018 – October 2018

- Architected ML-powered predictive models for personalized car rental recommendations, increasing conversion rates by 18% and average booking value by 25%
- Awarded 'Best Employee of the Month' twice for developing real-time price comparison and weather-based recommendation features
- Developed 20+ RESTful APIs using Node.js and Express.js, processing 100,000+ requests per minute with 99.99% uptime
- Implemented A/B testing framework informing product decisions affecting 200M+ travelers annually

Associate Consultant | Sapient India | January 2016 – May 2018

- Developed full-stack enterprise applications using Java Spring Boot, Node.js, and React, serving 50,000+ daily active users
- Integrated AI capabilities using WIT.ai for NLP in customer service chatbot, reducing support ticket volume by 35%
- Designed microservices architecture for banking client, breaking monolithic application into 15+ independent services
- Optimized database queries achieving 70% performance improvement for Oracle and MongoDB systems
- Wrote and reviewed design documents explaining overall architecture, framework, and high-level design for developers

EDUCATION

Master of Computer Applications (MCA) | Birla Institute of Technology (BIT), Mesra | 2016

Specialized in Distributed Systems, Computer Networks, and Database Management. Master's Thesis: 'Implementing Consensus Mechanisms in Distributed Ledger Technologies'. Published 2 research papers in international conferences on distributed computing.

CERTIFICATIONS

Blockchain & Web3: Hedera Improvement Proposal (HIP) Author & Approver, Ethereum Smart Contract Security Auditor, Certified Hyperledger Fabric Developer, Solana Certified Developer, Cardano Plutus Pioneer Program Graduate

Cloud & Architecture: AWS Certified Solutions Architect - Professional, Kubernetes Certified Administrator (CKA), Docker Certified Associate, Microsoft Azure Solutions Architect

Open Source Contributions: Core contributor to Urbit decentralized operating system (Identity and Journal Lead), Recipient of $500K Urbit Foundation grant, Active contributor to Web3.js, Ethers.js, and Solana Web3 JavaScript libraries

PROJECTS & PUBLICATIONS

Published 5+ technical articles on blockchain development reaching 50,000+ developers